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Impact of Ge clustering on the thermal conductivity of SiGe nanowires: atomistic simulation study

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Using non-equilibrium molecular dynamics simulations, we demonstrate that the thermal conductivity of SiGe alloy nanowires is remarkably sensitive to inhomogeneous composition distributions. Specifically, the effects of Ge clustering on the thermal conductivity of SiGe nanowires are studied. The results showed that clustering Ge atoms can improve the thermal conductivity of SiGe alloy nanowires due to the reduction of random alloy scattering centers. When the number of Ge atoms in the nanocluster increases, the thermal conductivity of such nanowires grows monotonically compared with that of random alloy nanowires. To reveal the role of inhomogeneous Ge distributions on the thermal conductivity, we performed vibrational eigenmode analyses and found the remarkable delocalization of phonon modes after Ge clustering. Through such analyses, we found that the increase in thermal conductivity was correlated with the phonon delocalization in the SiGe nanowires, where stronger delocalization indicates a better thermal performance of the nanowires. Our results are helpful not only in understanding the clustering effects on heat transport but also in modulating the thermal conductivity of SiGe nanowires.

1 Introduction

Silicon and silicon–germanium (SiGe) based nanostructures continue to be the focus of tremendous investment due to their widespread integration in nano- and optoelectronics,[1,2] solar cells,[3,4] thermoelectrics,[5–7] *etc.* Among the SiGe nanomaterials, one-dimensional SiGe alloy nanowires (NWs) have aroused widespread interest because they can function as a building block for nanoscale electronics and photonic systems. Specifically, SiGe nanowires are promising materials for the engineering of photodetectors,[8] energy storage devices,[9] high-efficiency solar cells,[10] and thermoelectric elements.[11] For these applications, understanding the thermal transport properties of SiGe NWs is essential. In particular, nanoelectronic devices demand a high thermal conductivity and heat sinks to dissipate the heat from nanochips, while a low thermal conductivity is important for achieving a high thermoelectric figure of merit. Consequently, numerous studies have been carried out to investigate the thermal properties of SiGe NWs.

In comparison with pristine Si NWs, a dramatic decrease in the thermal conductivity of silicon–germanium NWs was revealed in both experiments[12,13] and simulations.[14] Wang *et al.*[15] demonstrated the dependence of the thermal conductivity of SiGe alloy nanowires on the nanowire diameter, alloy concentration, and temperature. Qu[16] reported that the effect of the cross-sectional geometry on the thermal conductivity of SiGe NWs is weaker than that of pure Si NWs. By applying an *ab initio* phonon Boltzmann transport equation approach, Li and Mingo[17] found a strong dependence of the thermal conductivity on the crystal growth in SiGe alloy NWs.

Although Si–Ge forms, at thermodynamic equilibrium, a continuous series of substitutional solids, $Si_{1-x}Ge_x$, it is now possible to grow SiGe nanowires with highly complex structures that have inhomogeneous composition distributions. Numerous techniques have been proposed for the synthesis of these nanowires, involving chemical vapor deposition,[18] metal-catalyzed vapor–liquid–solid growth technology,[19] and combined laser ablation and thermal evaporation.[20] In the case of SiGe alloy nanowires, it has been demonstrated that inhomogeneous composition distributions can be obtained during their thermal oxidation.[21] In this process, Ge atoms diffuse and agglomerate as nanometer-sized particles. Among the physical properties, the thermal transport properties of NWs with inhomogeneous composition distributions are critical for their application in electronics and thermoelectric devices. Although several studies have been reported recently about the thermal properties of SiGe alloy structures,[22–24] including nanowires,[25,26] the quantitative impact of the inhomogeneous composition on the thermal transport in such nanowires remains unclear.

Theoretically, various techniques based on density functional theory, the Boltzmann transport equation, the Monte

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Carlo method, and continuous elasticity theory[27–31] have been used to address different aspects of the thermal transport phenomena in nanostructures. Recently, a substantially accelerated approach was proposed to study the thermal conductivity in bulk and nanoscale materials.[32,33] Another powerful tool to calculate the thermal conductivities of nanomaterials is molecular dynamics (MD) simulations. This technique solves the many body problem on an atomic basis and makes no assumptions about the scattering mechanisms, anharmonicity, and surface processes.[34] The molecular dynamics method has been widely used and has proved to be good in studies on the phonon properties of semiconductor NWs.[35,36]

In this work, we present a study of the lattice thermal conductivity of SiGe alloy NWs with inhomogeneous composition distributions. The effects of Ge clustering on the thermal conductivity of nanowires are investigated in particular using non-equilibrium molecular dynamics. In addition, we discuss the influence of such effects on the vibrational properties, including the density of states and the phonon participation ratio.

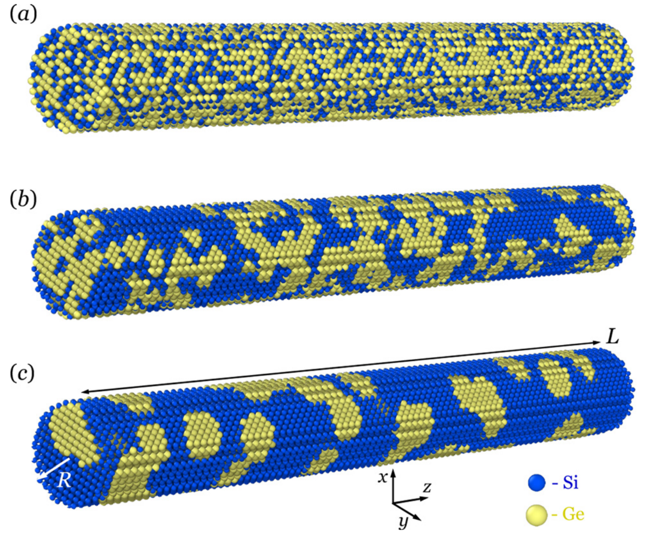

Fig. 1 Atomic configuration of the NWs used for calculation of the thermal conductivity: $Si_{0.5}Ge_{0.5}$ alloy nanowire (a); and Si + Ge_n nanowire with the number of Ge atoms in nanoclusters n, where n = 50 (b) and n = 500 (c).

2 Simulation details

The MD simulations for evaluating the thermal properties of SiGe nanowires with Ge clustering were carried out using the open-source software package LAMMPS.[37] The modeled nanowires were generated as follows. First, the Si nanowire was created using the inbuilt crystal structure generation algorithm present in LAMMPS. The crystal orientations [1 0 0], [0 1 0], and [0 0 1] were along the x, y, and z axes, respectively, as illustrated in Fig. 1. All atomic simulations performed in this study accounted for nanowires with a radius R = 2.7 nm and a length L = 49 nm. Then, the Si atoms were replaced randomly with Ge atoms in equal ratios to build a $Si_{0.5}Ge_{0.5}$ alloy nanowire. Finally, SiGe nanowires with inhomogeneous composition distributions were generated by rearranging the Ge atoms into nanoclusters using a special program code in Swift. In our notations, these nanowires are Si + Ge_n, where n is the number of Ge atoms in the nanoclusters. For all SiGe structures, the ratio of Si and Ge atoms was maintained while producing the nanoclusters. Nanowires with the number n of Ge atoms in nanoclusters from 10 to 500 were generated, as shown in Fig. 1. The modeled NWs were placed within an orthogonal simulation box that fits the length of the NW, but it is several times larger than its diameter in the transverse direction. The simulations were carried out under periodic boundary conditions.

The accuracy of the MD simulations is strongly related to the definition of the interatomic potential in a system. Many silicon potentials have been developed, such as the Stillinger–Weber potential,[38] the Tersoff potential,[39,40] and modified embedded-atom potential.[41] It is known that the thermal conductivity obtained with empiric potentials is higher than experimental values for bulk Si or individual Si NWs.[42–44] This discrepancy can be attributed to defects, impurities, surface roughness, and oxidation present in the experimental samples, which may impact adversely the thermal conductivity of the nanowires. By contrast, in ref. 45, it has been demonstrated that the thermal conductivity computed using the Tersoff potential agrees very well with that calculated using the full spectrum model, which uses the phonon dispersion inputs from *ab initio* calculations. Furthermore, it has been demonstrated that the Tersoff potential explains the structural, mechanical, and thermal properties of Si and SiGe nanostructures.[46–49] This indicates that the Tersoff potential is rather accurate to describe the phonon transport process in semiconductor NWs. Thus, we employed the Tersoff potential function in this work to calculate the Si–Si, Ge–Ge, and Si–Ge interatomic forces.

At the beginning of the simulation, energy minimization for each nanowire was performed using the conjugate gradient method. An initial velocity was assigned to each atom, which corresponds to the desired temperature. The structure was then equilibrated within 50 ps using the *NPT*-ensemble (at a constant number of particles N, pressure P, and temperature T) to allow the atoms to expand until nearly zero stress is achieved along the NW axis. The Nosé–Hoover thermostat was used to achieve a specific temperature and to stabilize it during the equilibration process. Newton's equations of motion were integrated using the velocity–Verlet algorithm, and the MD time step was set to 1.0 fs.

After system equilibration, the simulation was turned to the *NVE*-ensemble (at a constant number of particles N, volume V, and energy E) for 10 ns, and the atomic information was recorded every 1000 time steps to calculate the lattice thermal conductivity k. The reverse non-equilibrium molecular dynamics (RNEMD) method, which is also called the Müller–Plathe method,[50] was used to obtain the thermal conductivity of the Si + Ge_n NWs since this algorithm can accurately calculate the heat transfer properties of various materials. In the RNEMD method, each of the simulated nanowires is divided into several layers along its axis, and the outermost two layers are defined as the ''hot'' slab and the ''cold'' slab, respectively. The heat flux is then imposed across the simulated model along the axial direction by exchanging the velocity (energy) of the hottest atom in the cold slab with that of the coldest one in the hot slab. Energy exchange was completed once every 0.1 ps, and the heat flux in every swapping can be obtained according to the following equation:

$$Q_z = \frac{1}{2S}\frac{\mathrm{d}E}{\mathrm{d}t} = \frac{1}{2S\Delta t}\sum\left(\frac{m_\mathrm{h} v_\mathrm{h}^2}{2} - \frac{m_\mathrm{c} v_\mathrm{c}^2}{2}\right), \qquad (1)$$

where dE is the exchanged energy between the hot layer and the cold layer, which will lead to a temperature gradient after multiple time intervals dt, m_h and v_h present the mass and the velocity of the atom with the lowest kinetic energy in the hot slab, respectively, m_c and v_c present the mass and the velocity of the atom with the highest kinetic energy in the cold slab, respectively, Δt is the energy exchange time interval, and S is the cross-section of the model. The factor 2 arises from the periodicity.

When a steady state is reached, the temperature in the m-th slab can be evaluated by taking into account the kinetic energy of atoms according to the equipartition principle:

$$T_\mathrm{m} = \frac{1}{3N_\mathrm{m}k_\mathrm{B}}\left\langle \sum_{i=1}^{N_\mathrm{m}} m_i v_i^2 \right\rangle, \qquad (2)$$

where N_m is the number of atoms in the slab, m_i and v_i are the mass and velocity of the atom i, respectively, and k_B is the Boltzmann constant.

After reaching a steady state, the temperature profile $T(z)$ in the axial direction, which decreases from the center to both ends of the nanowire, can be measured. A typical temperature profile at the temperature T = 300 K is shown in Fig. 2. It can be found that the temperature of the middle layers between the hot slab and the cold slab can be fitted as a linear function, and these temperatures will be used to calculate the temperature gradient dT/dz. Finally, combined with the obtained heat flux and temperature gradient, the lattice thermal conductivity k of the NW can be calculated based on Fourier's law of heat conduction as

$$k = -\frac{Q_z}{\mathrm{d}T/\mathrm{d}z}. \qquad (3)$$

It should be noted that, due to the periodic boundary condition, the finite size effect exists in the calculated thermal conductivity when the nanowire length L is shorter than the phonon mean free path. Schelling *et al.*[51] showed that the size effect on the thermal conductivity can be approximated *via* the following relationship: $1/k \approx 2/L + 1/l_\infty$, where l_∞ represents the phonon mean free path for an infinite system. In this study, however, the length effect is not included in the MD computations since our purpose is to compare the trend of the relative change in the thermal conductivity after Ge clustering rather than to obtain the absolute value.

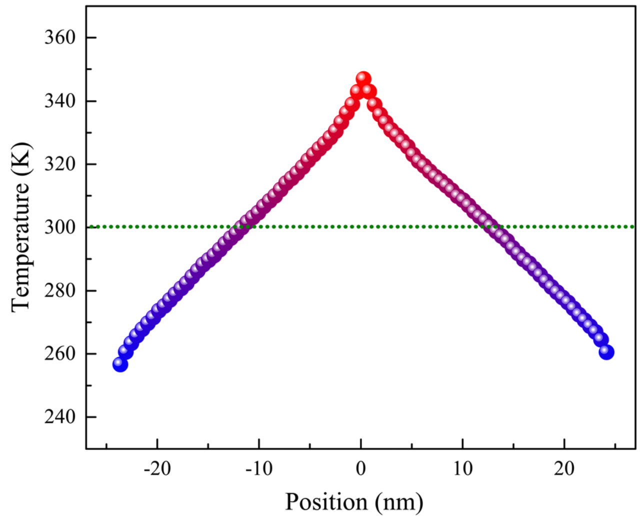

Fig. 2 Temperature profile in the Si + Ge_{50} nanowire at an average temperature of 300 K.

Due to the random nature of nanocluster formation, the thermal conductivities discussed here were obtained averaging over 2–3 independently generated configurations with the same number of Ge atoms in the nanoclusters. To obtain an ensemble average of the thermal conductivity at a given temperature, we performed 3 independent simulations for each nanowire, starting with different initial velocity distributions. The highest and lowest values of the thermal conductivity were excluded from the statistics. We used the standard error of the mean to represent the average and the error bars in the figures.

To gain more insight into the underlying physical mechanisms of thermal conductivity in nanowires, the vibrational density of states (DOS) $g(\omega)$ is analyzed using the Fourier transform of the velocity autocorrelation function,[52,53] in which $g(\omega)$ is written as

$$g(\omega) = \int \mathrm{e}^{i\omega t}\frac{\langle v_j(t)v_j(0)\rangle}{\langle v_j(0)v_j(0)\rangle}\mathrm{d}t, \qquad (4)$$

where v_j denotes the atom velocity at time t. Here, the summation is taken over all Si and Ge atoms in the system. In this MD simulation, the time interval was small enough to cover the maximum allowed lattice vibration frequency, and the velocities were recorded at every time step. The averaging was performed over 5.5 ps after the system reached equilibrium.

Finally, to interpret the thermal conductivity results, the phonon participation ratio was calculated using the quasi-harmonic Green–Kubo (QHGK) method.[54] These calculations were carried out using the open-source lattice dynamics calculator kALDo.[55] The QHGK method does not require any assumption of the nature of the heat carriers, and provides per-mode thermal conductivity estimations. Practically, the second- and third-order force constants were calculated using

LAMMPS before being imported into kALDo. All estimations in kALDo were performed using classical statistics.

3 Results and discussion

In order to confirm the accuracy of our MD simulations, we compared the room-temperature thermal conductivity of the $Si_{0.5}Ge_{0.5}$ alloy nanowire computed using the Tersoff potential with experimental data. Based on the NEMD results, we found that the thermal conductivity of the $Si_{0.5}Ge_{0.5}$ alloy NW at T = 300 K is 2.2 W m^{-1} K^{-1}, which is in good agreement with the experimental values of 2–4 W m^{-1} K^{-1}, obtained using various methods.[56,57] This suggests that the Tersoff potential is suitable for our NEMD simulation of SiGe NWs.

Following the computational model described in the previous section, the lattice thermal conductivity k of the Si + Ge_n NWs with Ge nanoclusters was evaluated at the temperatures of interest. Fig. 3 shows the temperature dependence of the thermal conductivities of the modeled NWs with different numbers n of Ge atoms in nanoclusters. The thermal conductivities of pristine Si, Ge, and $Si_{0.5}Ge_{0.5}$ alloy nanowires are shown as a reference.

As one can see, the thermal conductivity of the pristine Si and Ge nanowires decreases with temperature. For example, when temperature was varied from 300 K to 1000 K, the thermal conductivity of the Si NW was reduced by ~45%, from ~22.4 W m^{-1} K^{-1} to ~12.3 W m^{-1} K^{-1}. A similar decrease in thermal conductivity was also observed in previous studies for Si NWs.[58] By contrast, the thermal conductivity of the $Si_{0.5}Ge_{0.5}$ alloy nanowire has a considerably lower value and remains constant at $k \approx 2.2$ W m^{-1} K^{-1} throughout the entire temperature range studied here, analogous to its bulk counterpart.[22] Approximation of the $k(T)$ dependence using the power law $k = A{\cdot}T^{-b}$ gives $b \approx 0.5$ (0.4) for the Si (Ge) nanowire and $b \approx 0$ for the $Si_{0.5}Ge_{0.5}$ alloy nanowire. These results reflect the different natures of the dominant scattering mechanisms. In the context of pristine Si and Ge nanowires, the principal reason for the reduction in thermal conductivity is the coexistence of phonon–phonon, and surface scattering contributions, as described earlier.[58] However, in the case of the $Si_{0.5}Ge_{0.5}$ alloy NW, the temperature independence is believed to originate from the presence of non-propagating vibrational modes because of the localization of most of the high-frequency phonons around the impurity.[14] Finally, the thermal conductivity of Si + Ge_n NWs decreases slowly with temperature for a small number n of Ge atoms in nanoclusters, whereas this dependence becomes faster as n increases. In particular, by approximating the $k(T)$ dependence using a power law, we obtained $b \approx 0.05$, $b \approx 0.06$, and $b \approx 0.09$ for Si + Ge_n nanowires with the number n of Ge atoms in the nanoclusters as n = 10, n = 100, and n = 500, respectively. These results demonstrate that the thermal conductivity of SiGe alloy nanowires can be sensitive to the local clustering of the Ge atoms, and that Ge clustering leads to a change in the dominant phonon scattering mechanism.

Furthermore, local clustering not only changes the qualitative $k(T)$ dependence but also increases the thermal conductivity of the Si + Ge_n NW at a given temperature (Fig. 4). For example, the room-temperature thermal conductivity was increased from ~2.2 W m^{-1} K^{-1} for the $Si_{0.5}Ge_{0.5}$ alloy NW to ~4.1 W m^{-1} K^{-1} for the Si + Ge_{500} nanowire. According to the analysis above, increasing the Ge nanocluster size in the Si + Ge_n nanowires has two opposite effects on the phonon scattering. On the one hand, the Ge clusters cause the appearance of additional phonon scattering at the Si nanowire/Ge nanocluster interface, which results in a decrease in thermal conductivity. On the other hand, such Ge clustering will reduce the number of scattering centers, thus offering a mechanism by which increase the thermal conductivity compared with when Ge atoms are homogeneously distributed in the $Si_{0.5}Ge_{0.5}$

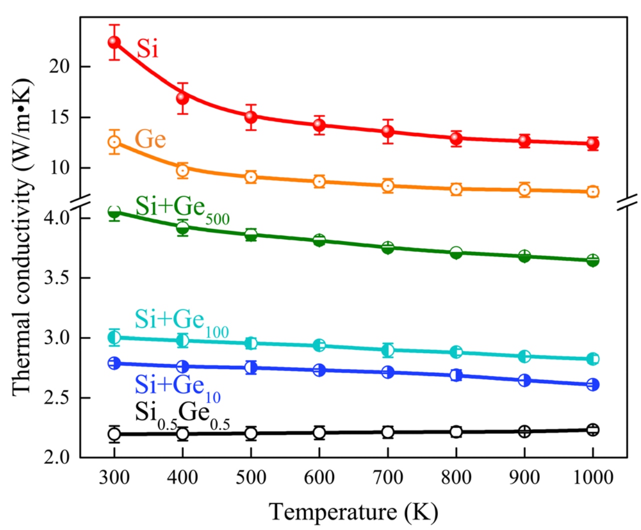

Fig. 3 Temperature dependence of the thermal conductivity of Si + Ge_n nanowires with the number of Ge atoms in the nanoclusters n = 10, n = 100, and n = 500, in comparison with the $Si_{0.5}Ge_{0.5}$ alloy nanowire and pristine Si and Ge nanowires.

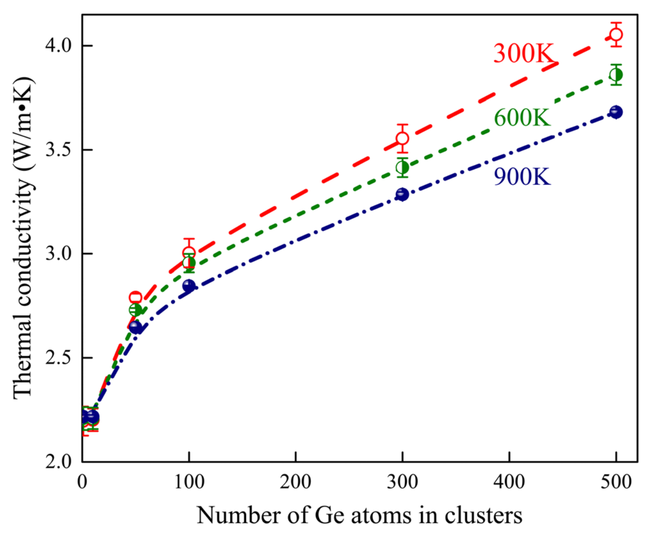

Fig. 4 Thermal conductivity of Si + Ge_n NWs as a function of the number n of Ge atoms in the nanoclusters at different temperatures: T = 300 K, T = 600 K, and T = 900 K.

matrix. The results presented in Fig. 4 show that the latter effect is dominant.

To further investigate the clustering effect, we calculated the vibrational DOS of the NWs. As can be seen in Fig. 5, the general shape of the vibrational DOS for the pristine Si and Ge NWs agrees with the previous theoretical studies and becomes quite similar to that of the bulk.[43,59] In the case of the Si (Ge) NW, the vibrational modes between around 5 THz and 10.5 THz (3.5 THz and 6.5 THz) can be roughly assigned to the transverse and longitudinal acoustic bands, while the optical bands are located roughly from 12 THz (7 THz) to 15 THz (9 THz). In the case of the $Si_{0.5}Ge_{0.5}$ alloy nanowire, the vibrational DOS consists of three peaks: the bulk Ge optical phonon band around 9 THz, the bulk Si optical phonon band around 15 THz, and the intermediate peak with frequencies around 12.5 THz, which has been attributed to the vibration of the Si–Ge bonds. In addition, a wide transverse acoustic band with frequencies 2–6 THz can also be seen. According to Fig. 5, Ge clustering leads to significant damping of the Si–Ge vibration band, and the DOS of the Si + Ge_n nanowire contains only optical bands corresponding to Si–Si and Ge–Ge bond vibrations. Moreover, Fig. 5 shows the higher intensity of the acoustic band above 5.0 THz compared with the $Si_{0.5}Ge_{0.5}$ alloy nanowire (inset in Fig. 5). Since low-frequency acoustic vibrations conduct heat more efficiently, this effect can explain the increase in thermal conductivity after Ge clustering.

To better understand the impact of Ge clustering on the thermal conductivity, we carried out a vibrational eigenmode analysis on different SiGe nanowires before and after clustering using the kALDo package. The mode localization can be qualitatively assessed *via* the participation ratio P_λ(PR), which is defined for eigenmode λ as

$$P_\lambda^{-1} = N\sum_i\left(\sum_\alpha \varepsilon^*_{i\alpha,\lambda}\varepsilon_{i\alpha,\lambda}\right)^2, \tag{5}$$

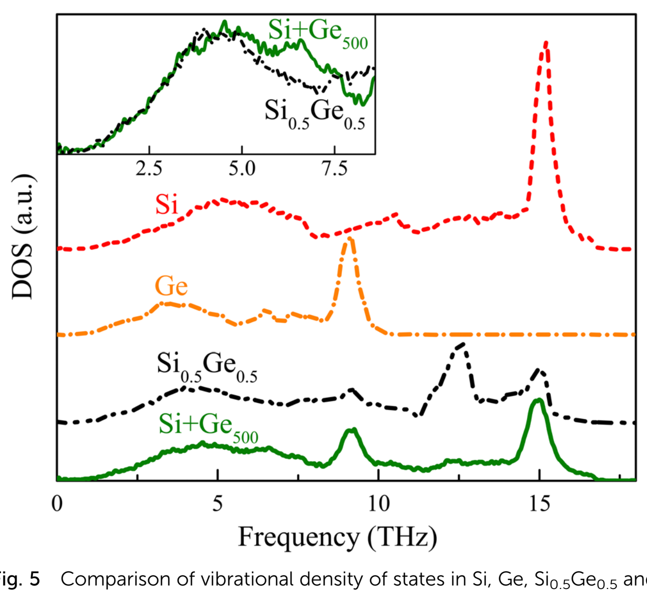

Fig. 5 Comparison of vibrational density of states in Si, Ge, $Si_{0.5}Ge_{0.5}$ and Si + Ge_{500} nanowires. The inset shows the vibrational DOS in the low-frequency region in the $Si_{0.5}Ge_{0.5}$ and Si + Ge_{500} nanowires.

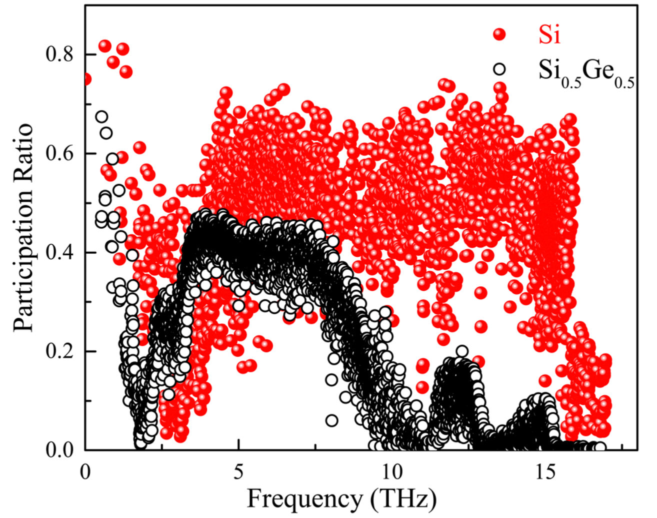

Fig. 6 Participation ratio for pristine Si NWs and $Si_{0.5}Ge_{0.5}$ alloy NWs.

where i sums over all atoms, N is the total number of atoms, α is a Cartesian direction and sums over x, y, z, and $\varepsilon_{i\alpha,\lambda}$ is the vibrational eigenvector component corresponding to the λ-th normal mode. The participation ratio P_λ indicates the fraction of atoms participating in a given vibrational mode and lies between 1 and $1/N$; that is, $P_\lambda = 1$ if all atoms vibrate with equal amplitude while $P_\lambda = 1/N$ for a mode involving only one atom. Typically delocalized modes are characterized as $P_\lambda \approx 0.5$–0.6.

The participation ratios of each vibrational mode for pristine Si NWs and $Si_{0.5}Ge_{0.5}$ alloy NWs are compared in Fig. 6. One can find that for the pristine Si NWs, the participation ratio mostly ranges from about 0.5 to almost 0.8, characterizing extended modes. These results are consistent with the MD results obtained by Xie *et al.*[60] However, we can observe distinct features of heat transport suppression in the $Si_{0.5}Ge_{0.5}$ alloy NWs, compared with pristine Si counterparts, through localization of the vibrational modes due to alloy scattering. It is evident in Fig. 6 that for the $Si_{0.5}Ge_{0.5}$ alloy NWs, the vibrational modes from 10 THz to 15 THz corresponding to Ge–Ge, Si–Ge, and Si–Si optical vibrations tend to be localized. Also, it was found that the PR value of acoustic vibrations around 3–7 THz decreases substantially in the $Si_{0.5}Ge_{0.5}$ NWs compared with the case of pristine Si NWs. For bulk crystals, due to the low group velocity of the optical phonons, their contributions to the thermal conductivity are very small, so that they are usually neglected. However, a recent study has reported that optical phonons can contribute over 20% to the thermal conductivity of nanostructures at room temperature.[61] Therefore, the alloying in SiGe NWs results in the remarkable localization effect of acoustic and optical phonons, thus leading to the reduction of thermal conductivity compared with the pristine Si or Ge nanowires.

For the $Si_{0.5}Ge_{0.5}$ alloy NWs with random Ge distributions, most of the PR values are below 0.5, implying that these phonon modes are localized. After Ge clustering is introduced (Fig. 7), there is a clear increase in the participation ratio for

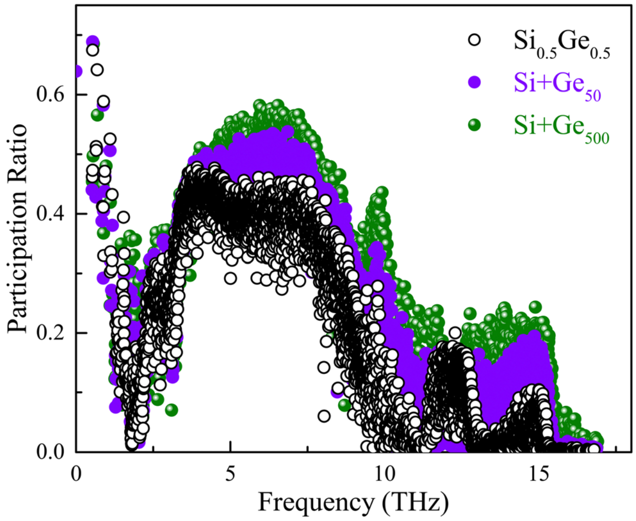

Fig. 7 Participation ratio for $Si_{0.5}Ge_{0.5}$ alloy NWs and Si + Ge_n NWs with n = 50 and n = 500.

both acoustic and optical vibrations in the Si + Ge_n NWs. Moreover, the PR values for the acoustic vibrations in the Si + Ge_n NWs with Ge nanoclusters are above 0.5, implying the delocalized modes. In addition, there is a clear increase in the participation ratios when the number of Ge atoms in the nanoclusters changes from n = 50 to n = 500, which indicates that larger clustering can enhance phonon delocalization.

In Fig. 8, the mean value of all mode participation ratios *versus* the number of Ge atoms in the nanoclusters for the Si + Ge_n NWs is shown. With the increase in Ge clustering, the mean participation ratio increases monotonically, indicating delocalization due to reduced random alloy scattering, which corresponds to high thermal conductivity. The contribution to thermal transport mainly comes from delocalized modes rather than localized modes. Therefore, these analyses show that the increase in Ge clustering induces a greater degree of phonon

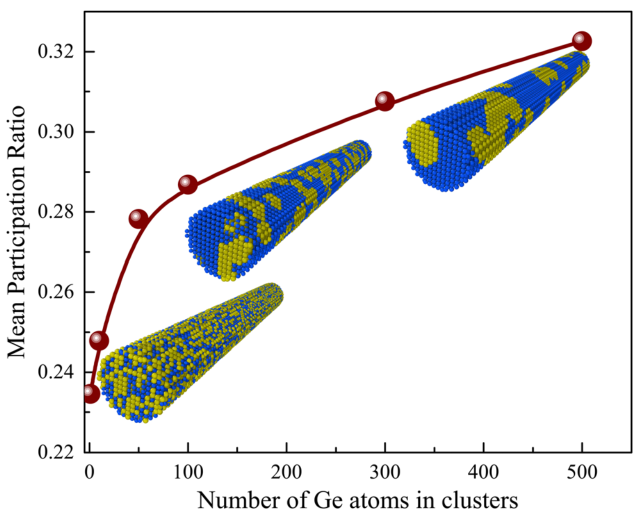

Fig. 8 Mean participation ratio in the Si + Ge_n NWs as a function of the number of Ge atoms in the nanoclusters.

delocalization in the Si + Ge_n NWs, causing the high thermal conductivity. The dependence of the mean participation ratios on the number of Ge atoms in the nanoclusters is consistent with the changes in thermal conductivity shown in Fig. 4. Therefore, the thermal conductivity of the SiGe NWs can be manipulated by varying the clustering.

4 Conclusions

In summary, using molecular dynamics simulations, we demonstrate that the thermal conductivity of SiGe alloy nanowires can be effectively tuned *via* the Ge clustering. Specifically, we found that the thermal conductivity increases monotonically as the clustering is extended. Our analysis shows the delocalization of phonon modes in the SiGe nanowires with Ge clustering compared with the random alloy nanowires, which turns out to be due to the reduced number of scattering centers. This fundamental understanding will provide some guidance on how to modify the SiGe alloy NWs to enhance their thermal properties.

Author contributions

V. Kuryliuk: conceptualization, methodology, software, formal analysis, investigation, writing – original draft, writing – review and editing O. Tyvonovych: methodology, software, formal analysis, investigation, resources, visualization, writing – original draft, writing – review and editing. S. Semchuk: investigation, data curation, formal analysis, writing – original draft, writing – review and editing.

Conflicts of interest

There are no conflicts to declare.

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